LITERATURE REVIEW

Presented by:
Group 5
Team Members:
Adeeba Abdul Vahid (KNP23CS011)
Avani P (LKNP23CS114)
Thara A (LKNP23CS117)
Guide Name : Mr. Anwar Sadath

Base Paper Name:

A Framework to Make Voting System Transparent Using Blockchain Technology

Authors: M. S. Farooq, U. Iftikhar, A. Khelifi

Year of Publication: 2022

How It Works:

Voters connect through a mobile/web dApp linked to a blockchain-based Voting Management System (VMS). After a national-ID check confirms eligibility and that the voter has not already voted, the ballot is turned into a transaction, mined, and added as a new block; a Chain Security Algorithm re-checks the chain's integrity every time a block is added.

Technologies Used:

Blockchain, Smart Contracts, flexible/plug-and-play Consensus Algorithms (default Proof-of-Work), Chain Security Algorithm, UTXO (Unspent Transaction Output) model, SHA-256 cryptographic hashing.

Limitations:

- Depends on constant, reliable internet connectivity for every voter
- No biometric layer — eligibility relies only on the National ID check
- 51% attack is only mitigated by pre-selected/monitored miners, not eliminated structurally
- Only evaluated through simulation (Remix/Solidity), not a real-world election

Conclusion:

The proposed VMS keeps response time and block-chain growth manageable even at 6,000,000 votes, showing that a flexible-consensus, Chain-Security-Algorithm-backed blockchain can deliver a transparent, tamper-proof voting system at large scale.

Base Paper Name:

Blockchain Integration With Multimodal Biometric Authentication System for Secure Smart Verifiable Electronic Voting System

Authors: L. A. Ajao, B. U. Umar, H. O. Ohize, E. M. Dogo, E. Esenogho, M. Cameron

Year of Publication: 2025

How It Works:

An ARM Cortex-A76/Raspberry-Pi device captures a voter's face and fingerprint, enhances the images (histogram equalization, Hough transform, Gabor filter) and matches them with a deep CNN; an RFID tag adds a further eligibility check. Once verified, the vote and biodata are encrypted, hashed, and recorded as a transaction on an Ethereum blockchain using a Proof-of-Stake consensus.

Technologies Used:

ARM Cortex-A76 processor, Raspberry Pi, RFID module, Deep Convolutional Neural Network (DCNN), Hough transform, Gabor filter, Ethereum blockchain, Solidity smart contracts, Pallier homomorphic encryption.

Limitations:

- Requires dedicated biometric hardware (camera + fingerprint sensor) at each voting point
- Image-processing/DCNN pipeline is computationally heavy
- Validated only on one small university election (2,423 students), not a national election
- Every vote transaction incurs a gas fee on the Ethereum network

Conclusion:

The system achieved 1,424.5 TPS, a 7.02s mining time, 0% False Acceptance/Rejection Rate and a 100% True Acceptance Rate in the NACOMES-FUTMinna 2024 election — confirming a highly secure, tamper-resistant, and accurate biometric-blockchain voting pipeline.

Base Paper Name:

Trustworthy Electronic Voting Using Adjusted Blockchain Technology

Authors: B. Shahzad, J. Crowcroft

Year of Publication: 2019

How It Works:

A consortium (permissioned) blockchain is 'adjusted' for the short polling window: a Presiding Officer with a verified identity creates each block without heavy Proof-of-Work, votes are added to it as transactions throughout the day, and once polling closes the block is permanently sealed using the authors' BSJC (Basit Shahzad & Jon Crowcroft) proof-of-completeness hashing scheme.

Technologies Used:

Consortium blockchain, SHA-256 hashing, Merkle trees, custom block-creation and block-sealing algorithms, biometric voter authentication (conceptual), integration with NADRA's national ID database.

Limitations:

- Purely conceptual — no working prototype, dataset, or experimental results are presented
- Assumes uninterrupted connectivity and a digitally literate voter base
- A private/consortium blockchain reduces transparency for ordinary voters versus a public ledger
- The election commission still retains central control over the blockchain

Conclusion:

By replacing expensive Proof-of-Work/Stake/Burn with a lightweight block-creation-and-sealing approach, the adjusted blockchain fits the short, bursty 8–10 hour polling window, offering a more cost- and time-efficient, tamper-evident foundation for trustworthy electronic voting.

Base Paper Name:

Intelligent Gesture-Enhanced Blockchain Voting: A New Era of Secure and Accessible E-Voting

Authors: W. A. Mahmood, J. Waleed, A. R. Abbas, H. Alaskar, M. Altulyan, A. J. Hussain

Year of Publication: 2024

How It Works:

A camera tracks the voter's face (Dlib landmarks) and hand (CVZone), letting them 'press' keys on a virtual keyboard just by raising an eyebrow, moving their nose, or pinching a finger — with no physical touch needed. The same gestures are used first to log in and then to select a candidate; each completed vote is turned into a transaction and mined into a Proof-of-Work blockchain block.

Technologies Used:

Dlib (68-point facial landmark detection), CVZone/OpenCV hand-gesture tracking, Proof-of-Work blockchain with a genesis block, Python, Flask web application, a gesture-driven virtual touchless keyboard.

Limitations:

- Needs a good-quality camera and stable lighting to track gestures reliably
- Finger-gesture accuracy (88.7%) trails facial gestures such as eyebrow movement (~92%)
- Older users take noticeably longer to interact (about 5.4s more than younger users)
- Evaluated through simulation and questionnaires rather than a live public election

Conclusion:

The system achieved 100% precision, 99% recall, and a very low Equal Error Rate, with user satisfaction between 61% and 75.5% — showing that combining gesture recognition with blockchain can make voting significantly more accessible for elderly and disabled users while remaining secure.

Base Paper Name:

Blockchain Integration With Multimodal Biometric Authentication System for Secure Smart Verifiable Electronic Voting System

Authors: L. A. Ajao, B. U. Umar, H. O. Ohize, E. M. Dogo, E. Esenogho, M. Cameron

Year of Publication: 2025

How It Works:

(This is a duplicate upload of Paper 2's file.) The same SSVEVS design applies: an ARM Cortex-A76/Raspberry-Pi device captures face and fingerprint biometrics, matches them with a deep CNN, and records every encrypted vote on an Ethereum blockchain secured with Pallier homomorphic encryption.

Technologies Used:

ARM Cortex-A76 processor, Raspberry Pi, RFID module, Deep Convolutional Neural Network (DCNN), Hough transform, Gabor filter, Ethereum blockchain, Solidity smart contracts, Pallier homomorphic encryption.

Limitations:

- Requires dedicated biometric hardware (camera + fingerprint sensor) at each voting point
- Image-processing/DCNN pipeline is computationally heavy
- Validated only on one small university election (2,423 students), not a national election
- Every vote transaction incurs a gas fee on the Ethereum network

Conclusion:

Identical results to Paper 2: 1,424.5 TPS, 7.02s mining time, 0% FAR/FRR and 100% True Acceptance Rate — reconfirming the reliability of the biometric-blockchain voting pipeline.

Base Paper Name: Analysis of Blockchain Solutions for E-Voting: A Systematic Literature Review

Authors: A. Benabdallah, A. Audras, L. Coudert, N. El Madhoun, M. Badra **Year of Publication:** 2022

How It Works: Reviews and compares blockchain-based e-voting proposals (2010–2021) across the five phases of e-voting (initialization, identification, voting, counting, results), then benchmarks the surveyed applications against four thematic tables: voter authentication, vote encryption/hashing, resistance to attacks, and security properties.

Technologies Used: Ethereum & Hyperledger Fabric smart contracts, ECDSA public/private-key auth, biometric authentication, SHA-256, homomorphic encryption (Paillier/ElGamal), Zero-Knowledge Proofs, blind & ring signatures, PoW/PoS consensus.

Limitations:

- Blockchain cannot fully eliminate the need for a central authority to manage voter eligibility.
- Public blockchains struggle to scale to nationwide elections due to transaction throughput limits.
- Voter authentication remains a weak point — key theft/sale enables illegitimate vote casting.
- Coercion-resistance is rarely achieved; forgiveness (re-voting) is the best practical substitute.
- No surveyed system has been tested in a full-scale, real national election.

Conclusion: Blockchain enhances transparency, auditability and integrity of e-voting but is not a fully secure standalone solution for national elections; it is best positioned as a complement to existing voting infrastructure (e.g., verifying paper-vote tallies) rather than a full replacement.

Base Paper Name: Leveraging Blockchain Technology for Secure Online Voting Systems: A Comprehensive Review

Authors: Astha Sah, Ankit Kumar, Bharat Bhushan **Year of Publication:** 2025

How It Works: Surveys online voting system (OVS) architecture and blockchain fundamentals, then reviews how blockchain frameworks (Ethereum, Hyperledger) and tools (Ganache, Truffle, MetaMask) are combined with cryptographic techniques to build secure OVS, comparing them against traditional and e-voting systems and real government/commercial deployments.

Technologies Used: Blockchain (Bitcoin, ZCash, Ethereum, Exorum, Quorum), consensus protocols (PoW, PoS, DPoS, PBFT), hashing, blind signatures, ZKP, Ganache/Truffle/MetaMask/Geth toolchains.

Limitations:

- Scalability remains a serious concern — poor internet access limits rural adoption.
- Full decentralization is hard to achieve without sacrificing speed and security.
- Susceptible to hash-based, authenticity, and smart-contract-flaw cyberattacks.
- Real-world case studies (Australia, Russia) show only partial success and residual trust issues.
- User authentication and registration challenges persist across most implementations.

Conclusion: Blockchain-based voting is more secure, transparent, and cost-effective (≈87% fewer fraud cases, ~10% higher turnout in pilots) than traditional/e-voting systems, but scalability, decentralization, and authentication challenges must be resolved — future work should target IoT/biometric-based authentication for inclusive e-voting.

Base Paper Name: BP-Vot: Blockchain-Based e-Voting Using Smart Contracts, Differential Privacy, and Self-Sovereign Identities

Authors: Hamza Baniata, Giovanny Caluna **Year of Publication:** 2025

How It Works: Deploys a smart contract on Hyperledger Besu that anonymizes votes via a novel (k,ε)-differential-privacy scheme: a randomly chosen 'pivot' candidate has some votes statistically reassigned to other candidates, then approximates the true tally afterward. Self-Sovereign Identity (web3.0 wallets/DIDs) lets voters cast verifiable, pseudonymous votes without revealing their real identity to the contract.

Technologies Used: Ethereum smart contracts, Hyperledger Besu (cloud nodes in EU/USA), (k,ε)-differential privacy, Self-Sovereign Identity/DIDs & verifiable credentials (web3.0), Min-Max regression accuracy evaluation.

Limitations:

- Uses a single pivot candidate; privacy guarantees weaken slightly as candidate count grows.
- Tested only up to 3 BC nodes and 50k votes — larger node counts are known to raise latency.
- Relies on assumption that the election authority is honest regarding voter identity data.
- Adoption depends on future EU-wide rollout of standardized web3.0 digital identity wallets.

Conclusion: BP-Vot achieves 24% lower latency (~1 s/TX) than prior state-of-the-art and ≥98% vote-approximation accuracy while formally proving robustness against reconstruction attacks, demonstrating that differential privacy, smart contracts, and SSI can jointly deliver a GDPR-compliant, anonymized blockchain e-voting framework.

Base Paper Name: Toward Secure Electronic Voting: A Survey on E-Voting Systems and Attacks

Authors: Riccardo Barelli, Mario D'Onghia, Stefano Longari **Year of Publication:** 2025

How It Works: Defines fundamental (integrity, anonymity, eligibility, accuracy, verifiability, receipt-freeness, coercion-resistance, software independence) and secondary e-voting properties, surveys on-site, remote, and blockchain-based voter-verifiable systems, then catalogs real-world and theoretical attacks (ballot modification, stuffing, secrecy violation, jamming) with their mitigations.

Technologies Used: Mix networks/onion routing, homomorphic encryption, group/ring/blind signatures, zero-knowledge proofs, public vs. private blockchains, PoW/PoS, smart contracts (Helios, Belenios, Civitas, Ordinos, Kryvos, etc.).

Limitations:

- Blockchain-based systems struggle with scalability/latency at national election scale.
- Blockchain transparency can inadvertently de-anonymize voters, undermining vote secrecy.
- Coercion resistance remains largely unsolved for remote and blockchain-based systems.
- Most solutions don't address risks from compromised voter devices or insecure networks.
- Quantum computing threatens the cryptographic primitives many systems rely on.

Conclusion: Despite decades of research and many promising e-voting/blockchain proposals, no system fully satisfies all fundamental properties for high-stakes public elections; the authors call for device-independent verifiability, scalable blockchain architectures, secure-by-design network protections, and quantum-resistant cryptography before real-world deployment.

Base Paper Name: Blockchain-Based Secure Voting Mechanism Underlying 5G Network: A Smart Contract Approach

Authors: S. Chaudhary, S. Shah, R. Kakkar, R. Gupta, A. Alabdulatif, S. Tanwar, G. Sharma, P. N. Bokoro **Year of Publication:** 2023

How It Works: Deploys a Remix IDE smart contract (Election, addCandidate, authorize, vote, end functions) that registers/validates voters & candidates via an election commission, stores voting data cost-efficiently via IPFS linked to the blockchain, and transmits votes over a low-latency, high-reliability 5G wireless network to determine the winning candidate.

Technologies Used: Ethereum smart contracts (Remix IDE), IPFS for data storage, 5G wireless network (MATLAB 5G toolbox), Proof-of-Work consensus, Echidna fuzz-testing security tool.

Limitations:

- Changing a vote after casting is difficult since users are only allowed to vote once.
- Smart contracts for large populations may contain loopholes that are hard to fully verify.
- Third-party interference risk remains since contracts must still be written/approved by people.
- Voter devices remain vulnerable to malware that could alter or forge votes before submission.

Conclusion: The proposed blockchain+IPFS+5G mechanism lowers gas/storage costs versus traditional blockchain-only schemes, achieves lower bit-error-rate than 4G/LTE-A, and passed Echidna security testing with no detected vulnerabilities, offering a secure, reliable, and cost-efficient e-voting foundation for future real-time deployments.

Base Paper Name: Design of Blockchain-Based Electronic Election System Using Hyperledger: Case of Indonesia

Authors: Donny Seftyanto, Amiruddin Amiruddin, Arif Rahman Hakim

Year of Publication: 2019

How It Works:

Uses a multi-organization Hyperledger Fabric model, with each Indonesian province acting as a separate organization. Voters authenticate via an e-ID card and fingerprint, then cast votes using a one-time smartcard-enabled Voting Booth. Each vote becomes a blockchain transaction (a Ballot asset), and results are monitored in real time by KPPS officers and the public.

Technologies Used:

Hyperledger Fabric, Hyperledger Composer, Hyperledger Explorer; e-ID card & fingerprint readers; smartcards; business network archive (.cto / .js / .acl / .qry)

Limitations:

- Preliminary design only — not implemented or tested in a real election
- No performance, security, or scalability evaluation was conducted
- Admin Server retains significant centralized control over participants
- Requires substantial hardware (e-ID readers, smartcards) at every polling station

Conclusion: The proposed design can maximally implement blockchain to solve Indonesia's 2019 election problems (logistics, slow counting, inconsistent rules, recapitulation errors) and satisfies all secure-election requirements; real-world implementation and testing are left for future work.

Base Paper Name: Blockchain-Enabled E-Voting

Authors: Nir Kshetri, Jeffrey Voas

Year of Publication: 2018

How It Works:

Each voter receives a digital “wallet” with a credential and a single “coin” representing one vote. Casting a vote transfers the coin to a candidate's wallet, creating a permanent, tamper-evident ledger entry. Voters may change their vote before a preset deadline, and any manipulation is detectable by the peer-to-peer consensus network.

Technologies Used:

Public / permissioned blockchains; smart biometrics & real-time ID verification (Voatz); Ethereum smart contracts (Nasdaq, Moscow's Active Citizen)

Limitations:

- No full national, legally binding election has used BEV to date
- Blockchain technology remains immature — high software defect rates (especially Ethereum)
- Requires reliable broadband access and digital literacy
- Authentication/validation is energy-intensive and slow, limiting national-scale use
- Shifts power from electoral authorities, inviting political resistance

Conclusion: Blockchain can improve voter access and reduce fraud through tamper-proof, auditable records, but immaturity, scalability, and trust-building challenges mean BEV is not yet ready for national elections.

Base Paper Name: Blockchain-Based System for E-Voting Using Blind Signature Protocol

Authors: Julio César Pérez García, Abderrahim Benslimane, Samia Boutalbi

Year of Publication: 2021

How It Works:

Each ballot is split into two parts (a line's point P and slope λ) so that neither part alone reveals the vote. A Central Authority issues a blind signature on the encrypted ballot without seeing its content, breaking the link between voter identity and vote. The Blockchain Network validates and stores ballots; after voting closes, published private keys allow decryption and tallying.

Technologies Used:

Hyperledger Fabric v1.4.0 + Docker; ElGamal-based Blind Signature; Elliptic Curve Cryptography; SHA-256; PBFT consensus; MATLAB discrete-event simulation

Limitations:

- Evaluated only via simulation, not a real deployment
- Average voting time grows with voter count and can approach 120 seconds under congestion
- Outage probability rises sharply as transaction rate nears network capacity
- Central Authority remains a single point of trust for token issuance

Conclusion: The blind-signature blockchain scheme achieves full unlinkability plus verifiability, eligibility, fairness, and anonymity — outperforming compared designs — with low per-voter computation cost (~0.047 ms) and acceptable scalability.

Base Paper Name: E-Voting System Based on Blockchain Technology: A Survey

Authors: Abdullah Quzmar, Mohammad Qatawneh, Sarah Al-Maaitah

Year of Publication: 2021

How It Works:

Surveys and compares eight blockchain-based e-voting proposals, covering architectures such as Ethereum smart contracts, double-envelope encryption, leveled/hierarchical designs, Proof-of-Authority networks, homomorphic encryption, hash-and-signature recording, Shamir's Secret Sharing sidechains, and multi-agent systems — evaluating each for security and design trade-offs.

Technologies Used:

Ethereum & smart contracts; Go-Ethereum Proof-of-Authority; homomorphic encryption; SHA-256; AES/DES; Shamir's Secret Sharing; sidechains; multi-agent systems

Limitations:

- Most reviewed systems are drafts/prototypes, not real-world deployments
- Several designs still rely on a centralized electoral commission for key management
- Scalability, latency, and node-synchronization issues persist across designs
- No standardized blockchain protocol exists specifically for e-voting

Conclusion: Blockchain shows strong potential to improve e-voting security, privacy, and cost, but current literature lacks mature, well-maintained implementations — leaving clear gaps for future research.

Base Paper Name: Trust and Security in Critical Digital Infrastructures: A Conditional Analysis of Acceptance and Intention to Change in Electronic Voting Systems

Authors: Celio Castro Wermelinger, Ari Melo Mariano, João Mello da Silva

Year of Publication: 2026

How It Works:

Using PLS-SEM on survey data from 1,851 Brazilian voters, the study models a second-order "Overall Trust" construct (Perceived Security + Trust in Technology) alongside UTAUT factors (Performance/Effort Expectancy, Social Influence) to explain both Behavioral Intention to keep using e-voting machines and Intention to Change toward printed vote verification, then tests moderation by gender, age, education, and political orientation.

Technologies Used:

PLS-SEM (SmartPLS); UTAUT model; PLS-MGA multi-group analysis; 7-point Likert online survey; G*Power sample estimation

Limitations:

- Non-probabilistic online sample skews toward highly educated, male, older, digitally engaged voters
- Measurement invariance (MICOM) was not tested before group comparisons
- Intention to Change is explained less well ($R^2 = 0.588$) than Behavioral Intention ($R^2 = 0.848$)
- Effort Expectancy → Behavioral Intention and Social Influence → Intention to Change were not significant

Conclusion: Overall Trust is the dominant force behind both accepting e-voting machines and demanding printed vote verification (positively and negatively, respectively), and this trust dynamic varies significantly by gender, age, education, and political orientation — pointing to a need for audience-tailored transparency policies.

THANK YOU